



Anand Ram S

Greater Detroit Area, Michigan | 734 (881) 0808

anandramsathyanarayanan@gmail.com

<https://www.linkedin.com/in/anand-ram-s/>

EXPERIENCE

Siemens Digital Industries Software, Livonia, Michigan | FEB '18 - PRESENT

Application Engineer, Engineering & Consulting Services Division

Ford Bronco Driveline Controls

- Defined, developed and implemented advanced driveline controls algorithms from concept to production (full-product lifecycle) using System V-Cycle, including software architecture design, requirements capture, modeling, Design Verification Plan (DVP), code generation and testing (MIL/SIL)
- Designed & developed supervisory control software for effective torque management of 4x4 off-road capable All-Wheel-Drive (AWD) system including transfer case state estimation, clutch control and electronic locker controls
- Led client operations as Siemens Engineering Consultant for project planning and effective delivery of software products using Agile methodology

HMC Diesel Engine Controls

- Collaborated with global cross-functional teams to execute robust controls algorithms and plant models to improve diesel engine combustion sound quality
- Conducted Design of Experiments (DoE) and multi-attribute optimization to reduce in-vehicle testing time & cost using Amesim-Simulink co-simulation

GKN Driveline North America Inc., Auburn Hills, Michigan | MAY '17 - DEC '17

Vehicle Controls Intern, Embedded Software Controls Division

- Performed software requirements analysis and designed functional test cases for GM All-Wheel-Drive (AWD) Rear Drive Unit (RDU)
- Built auto-code and executed MIL/SIL/static model analysis to validate controls strategies using rapid prototyping tools for traction & stability control module

Ashok Leyland Ltd., Chennai, India | AUG '14 - JUNE '16

Deputy Manager, Technical Service & Strategy Division

- Collaborated with service, product development & reliability teams for addressing field failure issues for vehicle & chassis systems
- Spearheaded service strategy & analytics initiatives to develop models & logic for data-driven process & product improvements for over 500 dealerships across India
- Led training initiatives for over 150 service executives across India in dealership workshop process improvements to maximize revenue growth

Product Engineer, New Product Development Division

- Designed & implemented P-diagram, interface matrix, Quality Function Deployment (QFD), DFMEA, 3D Modeling & FEA to optimize Class 6 & 7 truck load body
- Applied Lean Six Sigma (LSS) principles to address load body structure issues such as floor sagging, column buckling and tail gate failure in Ashok Leyland BOSS model

PROJECTS

Control of hybrid pick-up truck with multi-mode power-split configuration for optimized fuel economy, University of Michigan, Ann Arbor

- Implemented multi-mode power-split configuration for Ford F-150 hybrid truck utilizing Toyota Prius architecture
- Developed Equivalent Consumption Minimization Strategy (ECMS) based hybrid controller for various drive cycles and applied dynamic programming to determine deviation from optimal operation zone

Modeling & simulation of adaptive shift scheduling system for automatic transmission, University of Michigan, Ann Arbor

- Developed an innovative adaptive shift scheduling algorithm based on generalized load concept for 6-speed automatic transmission for Ford F-150
- Reduced gear hunting and improved fuel economy for various drive cycles

EDUCATION

University of Michigan, Ann Arbor, MI

Master of Engineering | SEP '16 - DEC '17
Integrative Systems + Design (ISD),
Global Automotive & Manufacturing
Engineering | CGPA 3.7/4.0

Anna University, Chennai, India

Bachelor of Engineering | APR '10 - APR '14
Mechanical Engineering | CGPA 9.1/10.0

SKILLS

Software

Matlab
Simulink
Amesim
ATI Vision
Codeprover
Smartbear
GT Power
TargetLink
Solidworks
Ansys
Python
C/C++

Tools & Techniques

Model Based Design (MBD)
Model-in-the-loop Testing
Software-in-the-loop Testing
DVP&R
DFMEA
ISO 26262
3D CAD Modeling
CAE Analysis
Lean Six Sigma
Agile
Data Analytics
Project Management

AWARDS & LEADERSHIP

Ford Recognition Award | MAR '21

Ford Bronco driveline controls, Ford Motor Company, Michigan

Vice President | SEP '16 - DEC '17

Graduate Student Advisory Committee (GSAC), Integrative Systems + Design, University of Michigan, Ann Arbor

Design For Six Sigma Green Belt | JUNE '14

Ashok Leyland Ltd., New Product Quality, Chennai, India

Vice President | JUNE '13 - MAY '14

Society of Automotive Engineers (SAE) College Club, Anna University, Chennai

University Gold Medalist | Class of 2014

Rank 10 (among 7124 candidates), Bachelor of Engineering, Mechanical Engineering, Anna University, Chennai, India